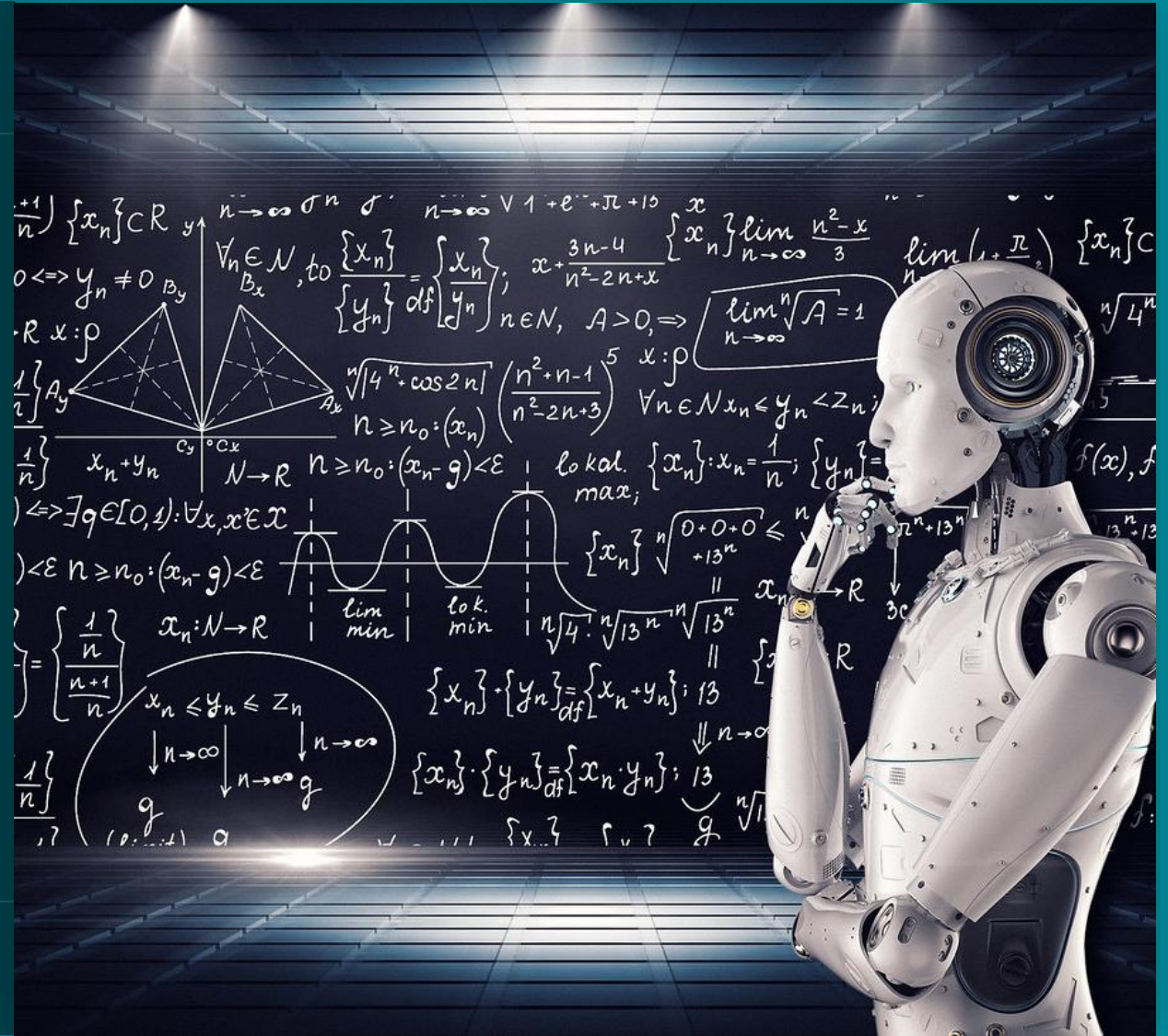


2nd year Medicine

Introduction to Machine Learning in Healthcare

Optional Course | Class 1

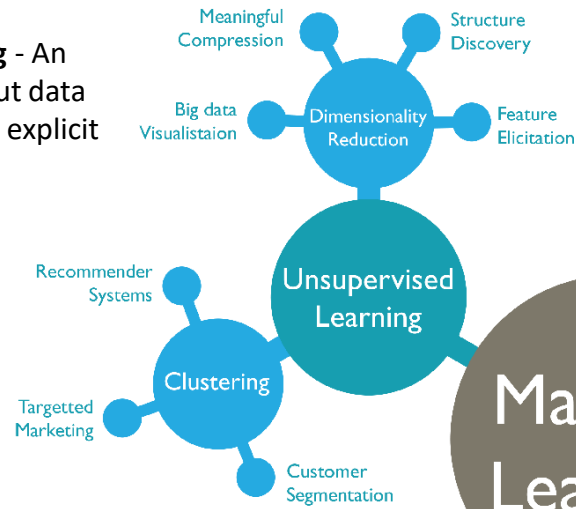


Quick definitions

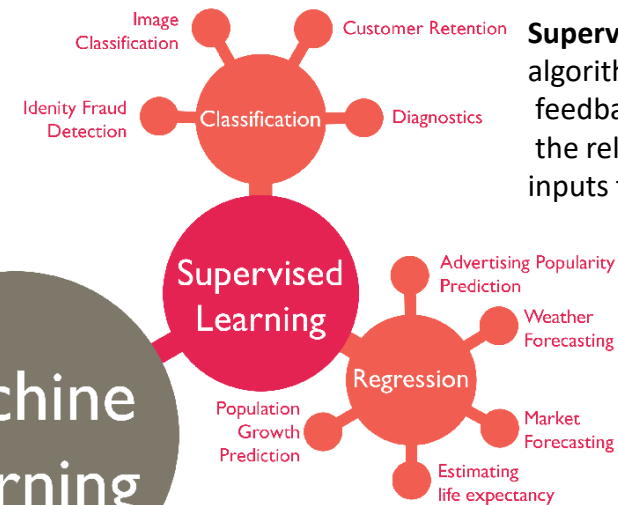
- **AI:** Making machines perform tasks requiring human intelligence
- **ML:** Algorithms learning patterns from data

Major types of ML algorithms

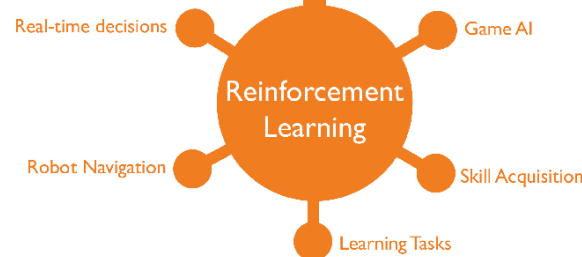
Unsupervised Learning - An algorithm explores input data without being given an explicit output variable.



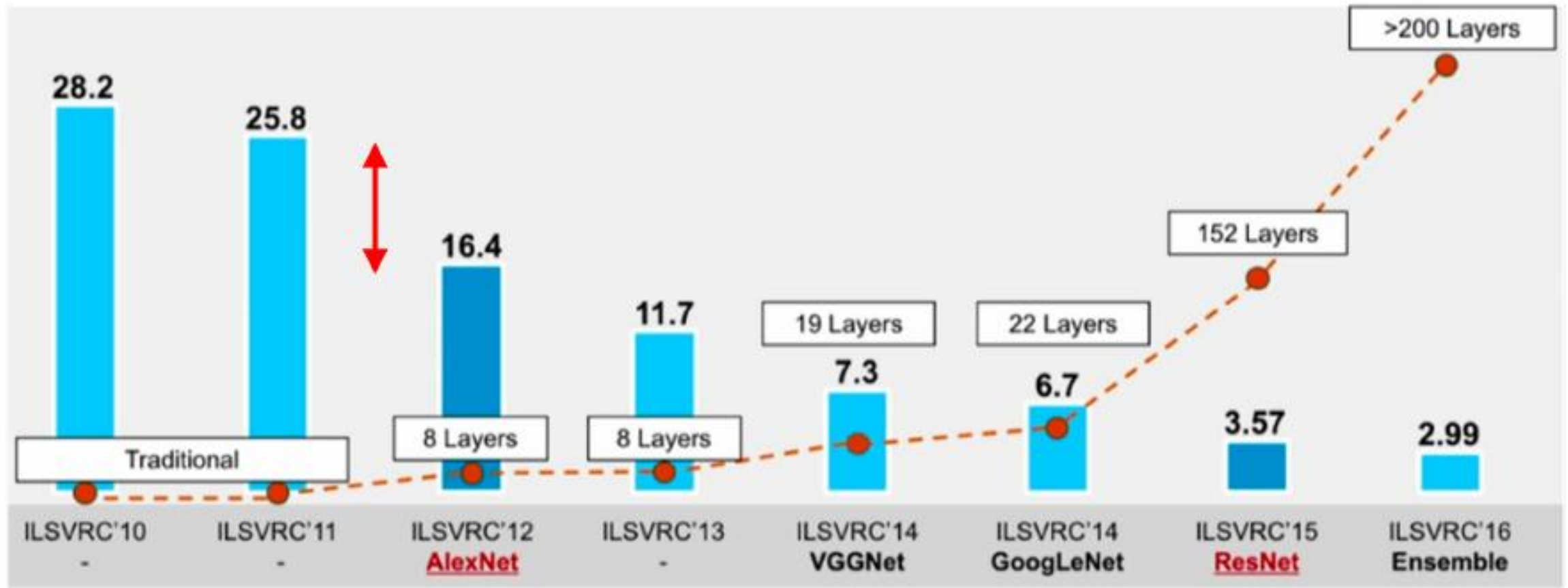
Supervised Learning - An algorithm uses training data and feedback from humans to learn the relationship of given inputs to a given output.



Reinforcement Learning - An algorithm learns to perform a task simply by trying to maximize rewards it receives for its actions.



Rise of Deep Learning: AlexNet



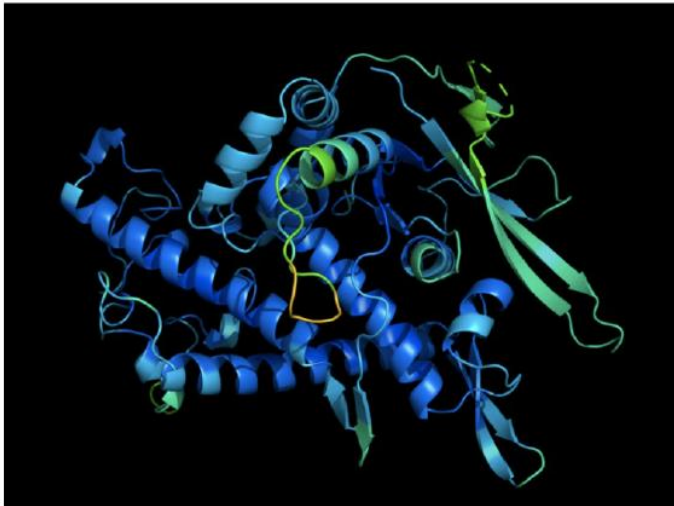
Accelerating rate of change

NEWS | 30 November 2020

'It will change everything': DeepMind's AI makes gigantic leap in solving protein structures

Google's deep-learning program for determining the 3D shapes of proteins stands to transform biology, say scientists.

Ewen Callaway



A protein's function is determined by its 3D shape. Credit: DeepMind

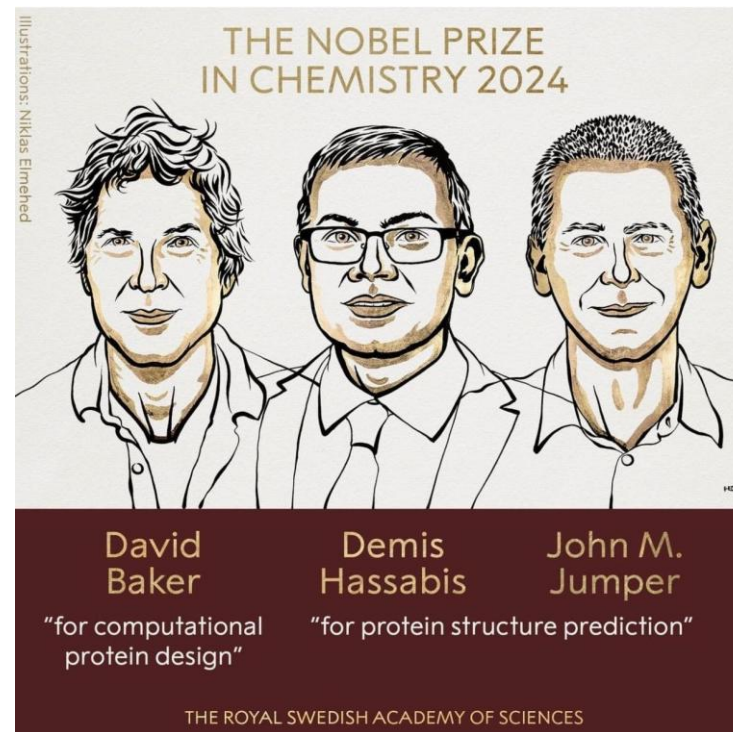
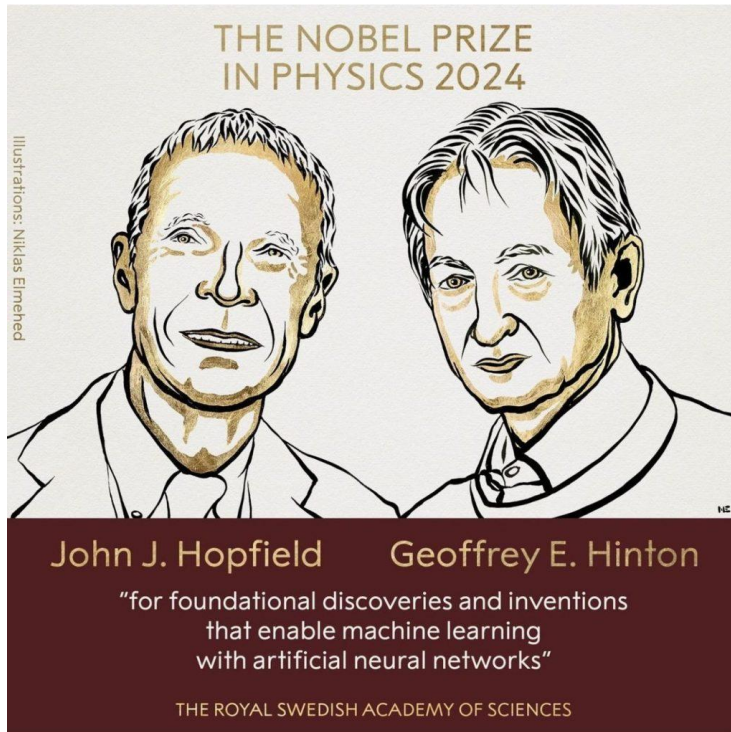
A robot wrote this entire article. Are you scared yet, human?

We asked GPT-3, OpenAI's powerful new language generator, to write an essay for us from scratch. The assignment? To convince us robots come in peace

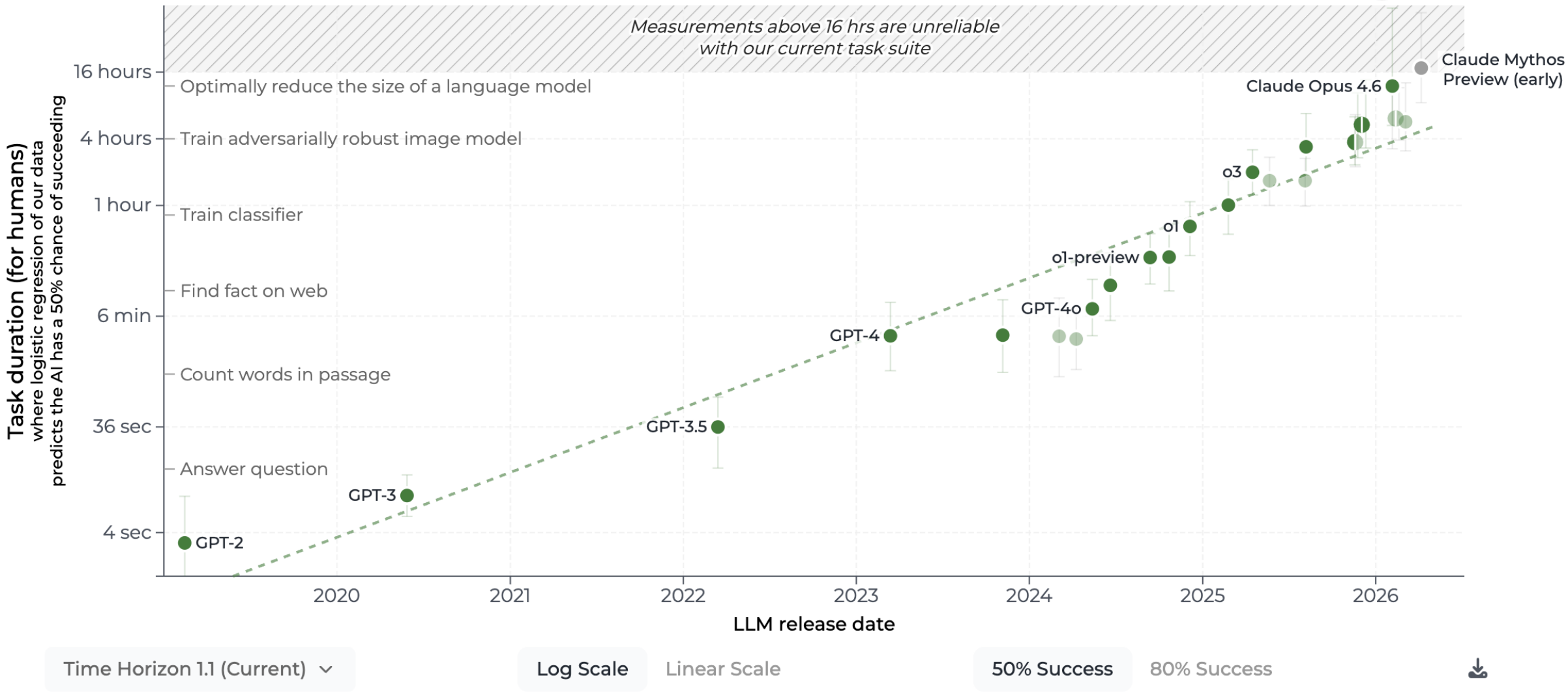
- For more about GPT-3 and how this essay was written and edited, please read our editor's note below



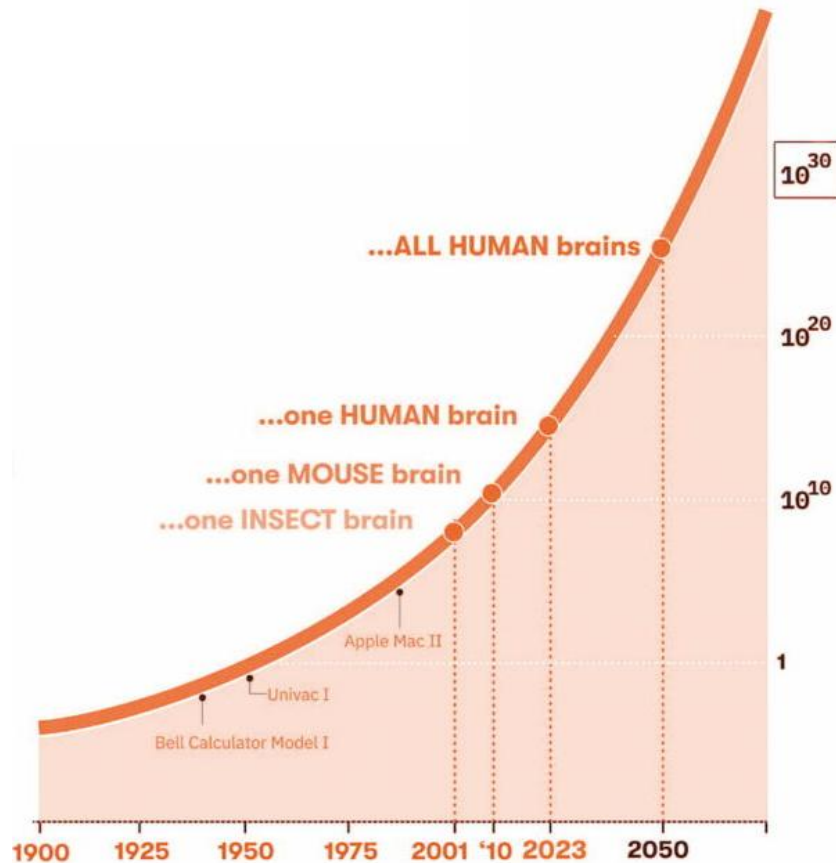
What is coming next?



Time horizon of software tasks
different LLMs can complete 50% of the time



Ray Kurzweil predictions



- **Artificial General Intelligence (AGI)** — an AI with human-level cognitive abilities — will be achieved by **2029**. This AGI will
 - Understand language, reason, and solve problems like a human
 - Pass a valid **Turing Test**, convincingly demonstrating human-like intelligence
- **Humans will start to merge with machines by the 2030s:**
 - By the **late 2020s to early 2030s**, he envisions **brain-computer interfaces** allowing us to connect our brains to the cloud
 - By **2045**, at the Singularity, he predicts a **full integration**—where human intelligence merges with artificial superintelligence, leading to radical changes in consciousness, identity, and capability

The healthcare challenge

Digital, data, and AI are opportunities to tackle our three key perennial healthcare challenges globally.



Increasing burden of chronic diseases

Rising prevalence of chronic conditions causing major pressure to healthcare systems globally



Inequitable outcomes and access

Wide and persistent inequities across the world as well as within countries



Resource constraints

Unsustainable cost growth and healthcare workforce crisis

1 year

more of a person's lifetime expected to be spent in ill health today vs. 10 years ago

4.5 b

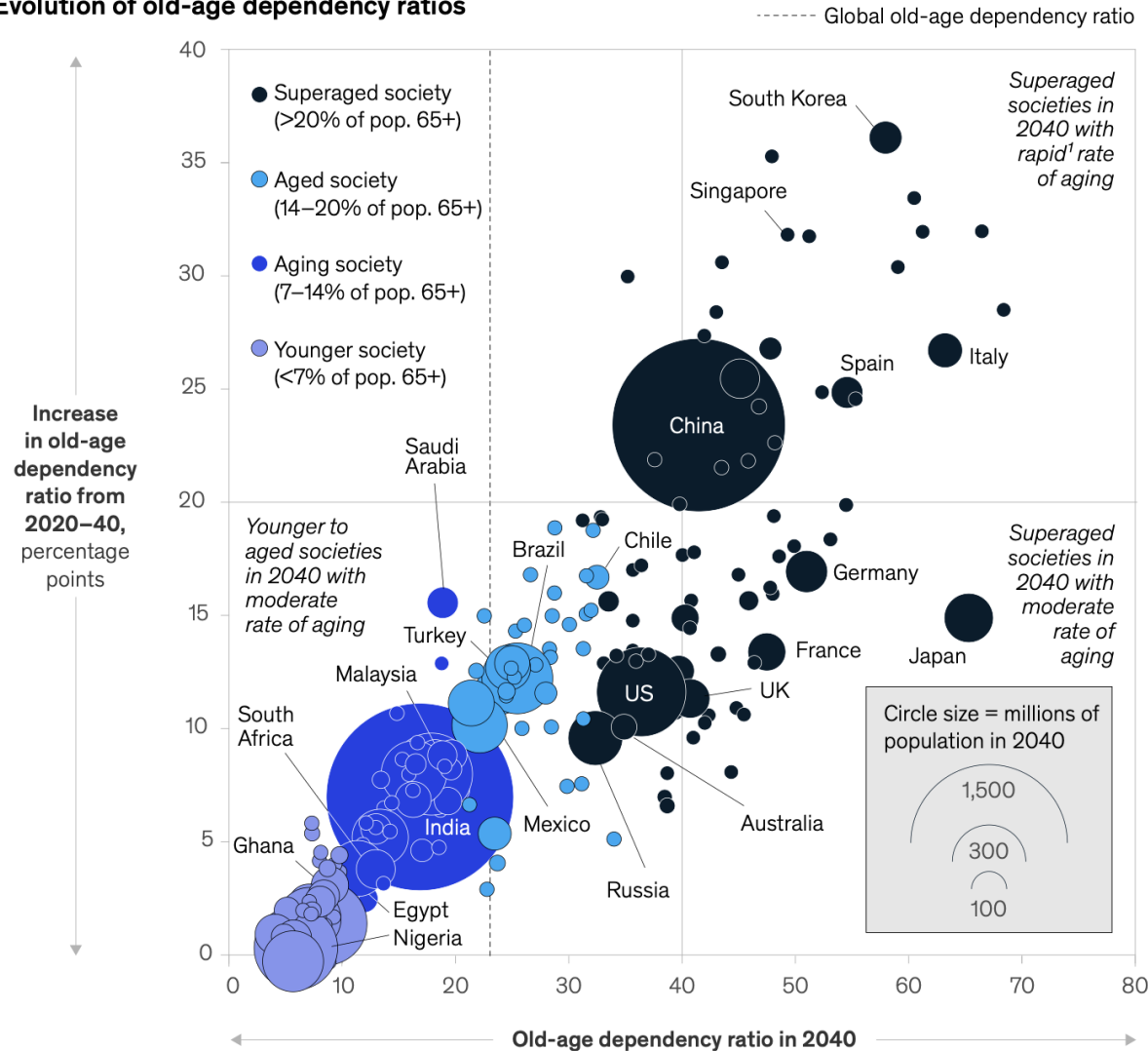
people currently lack access and coverage for essential healthcare services

10 m

health worker shortage expected by 2030; major gaps in developing countries

Some 'superaged' societies will see old-age dependency ratios increase substantially through 2040.

Evolution of old-age dependency ratios



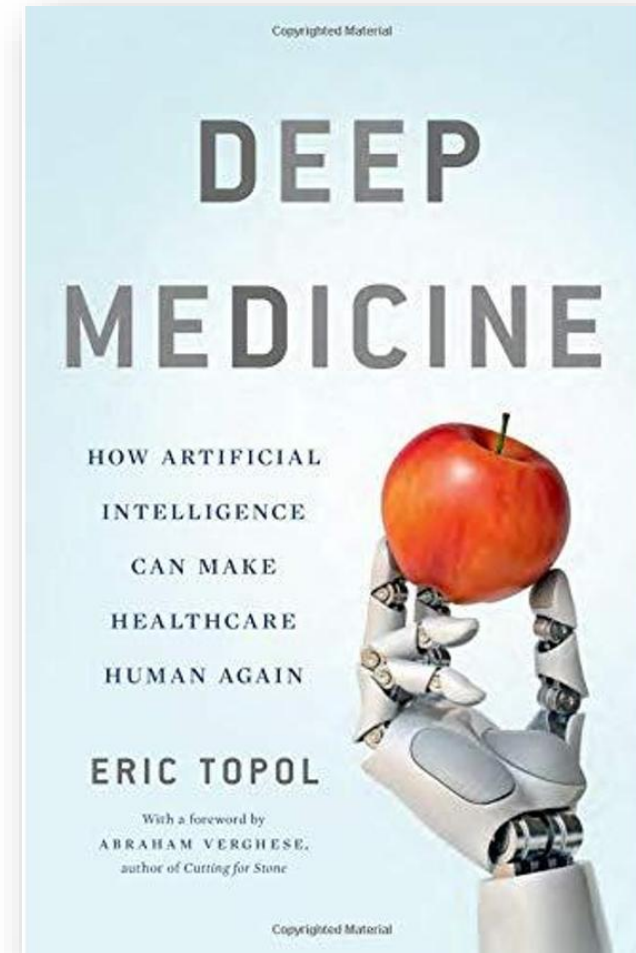
The most powerful force shaping healthcare today isn't technology....

It's aging.

¹Increase of old-age dependency ratio from 2020–40 is >20 percentage points.
Source: *World Population Prospects 2022*, United Nations

Why this matters for medicine?

- We train doctors for 10–15 years, for a world that won't be today's
- Using AI without judgment optimises for a world that will stop existing
- The bottleneck of medicine's future isn't technological, it's about training!



AI already matches or beats the doctor

- o1-preview gave 'perfect' diagnostic reasoning (R-IDEA scale) in 78 of 80 cases, vs 28/80 for specialists
- In ICU, intensivists 88% vs ChatGPT-4 85%: no significant difference ($p=0.25$)
- The capability is already here — and accelerating

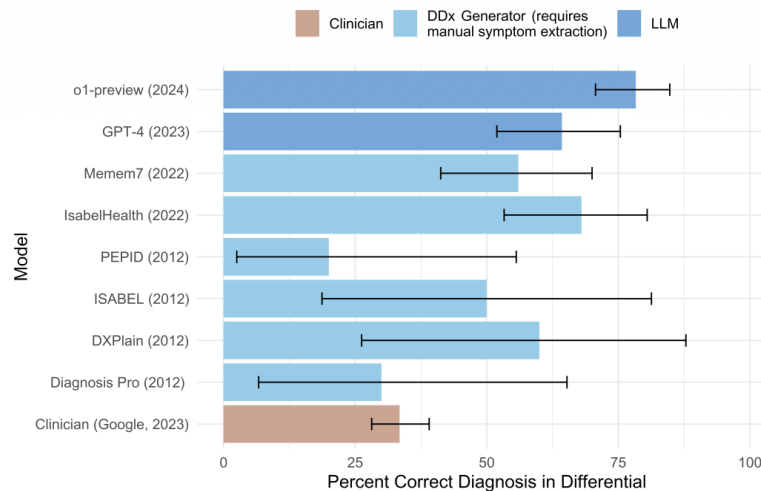


Fig.1. Performance of differential diagnosis generators and LLMs on NEJM clinicopathologic conferences (CPCs), 2012 to 2024. Bar plot showing the accuracy of including the correct diagnosis in the differential for differential diagnosis (DDx) generators and LLMs on the NEJM CPCs, sorted by year. Data for other LLMs or DDx generators were obtained from the literature (materials and methods). The 95% CIs were computed with a one-sample binomial test.

Original Research

Diagnostic Accuracy of a Large Language Model (ChatGPT-4) for Patients Admitted to a Community Hospital Medical Intensive Care Unit: A Retrospective Case Study

Journal of Intensive Care Medicine
2024, Vol. 41(5) 413-420
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DOI: 10.1177/08850666241368270
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Mary Ann Liebert
A Part of Sage

Jassimran Singh, MD¹, Rhea Bohra, MD¹, Vaibhavi Mukhtiar, MD¹, Warren Fernandes, MD¹, Charmi Bhanushali, MD¹, Rajaeaswaran Chinnamuthu, MD¹, Shihla Shireen Kanamgode, MD², June Ellis, RN³, and Eric Silverman, MD⁴

Abstract

Background: The future of artificial intelligence in medicine includes the use of machine learning and large language models to improve diagnostic accuracy, as a point-of-care tool, at the time of admission to an acute care hospital. The large language model, ChatGPT-4, has been shown to diagnose complex medical conditions with accuracies comparable to experienced clinicians, however, most published studies involved curated cases or examination-like questions and are not point-of-care. To test the hypothesis that ChatGPT-4 can make an accurate medical diagnosis using real-world medical cases and a convenient cut and paste strategy, we performed a retrospective case study involving critically ill patients admitted to a community hospital medical intensive care unit.

Methods: A redacted H&P was essentially cut and pasted into ChatGPT-4 with uniform instructions to make a leading diagnosis and a list of 5 possibilities as a differential diagnosis. All features that could be used to identify patients were removed to ensure privacy and HIPAA compliance. The ChatGPT-4 diagnoses were compared with critical care physician diagnoses using a blinded longitudinal chart review as the ground truth diagnosis.

Results: A total of 120 randomly selected cases were included in the study. The diagnostic accuracy was 88.3% for physicians and 85.0% for ChatGPT-4, with no significant difference by McNemar testing (p -value of 0.249). The agreement between physician diagnosis and ChatGPT-4 diagnosis was moderate, 0.57 (95% CI: 0.35–0.79), based on Cohen's kappa statistic.

Conclusion: These results suggest that ChatGPT-4 achieved diagnostic accuracy comparable to board certified physicians in the context of critically ill patients admitted to a community medical intensive care unit. Furthermore, the agreement was only moderate, suggesting that there may be complementary ways of combining the diagnostic acumen of physicians and ChatGPT-4 to improve overall accuracy. A prospective study would be necessary to determine if ChatGPT-4 could improve patient outcomes as a point-of-care tool at the time of admission.

We must learn to use AI well

- RCT, 50 doctors
 - With GPT-4 = 76% vs without = 74%
 - Not a significant difference
- And GPT-4 alone (~92%) beat both groups of doctors

→ The bottleneck is the human–AI interface and training, not the model



Original Investigation | Health Informatics

Large Language Model Influence on Diagnostic Reasoning A Randomized Clinical Trial

Ethan Goh, MBBS, MS; Robert Gallo, MD; Jason Hom, MD; Eric Strong, MD; Yingjie Weng, MHS; Hannah Kerman, MD; Joséphine A. Cool, MD; Zahir Kanjee, MD, MPH; Andrew S. Parsons, MD, MPH; Neera Ahuja, MD; Eric Horvitz, MD, PhD; Daniel Yang, MD; Arnold Milstein, MD; Andrew P. J. Olson, MD; Adam Rodman, MD, MPH; Jonathan H. Chen, MD, PhD

Abstract

IMPORTANCE Large language models (LLMs) have shown promise in their performance on both multiple-choice and open-ended medical reasoning examinations, but it remains unknown whether the use of such tools improves physician diagnostic reasoning.

OBJECTIVE To assess the effect of an LLM on physicians' diagnostic reasoning compared with conventional resources.

Key Points

Risk 1 — Automation bias & deskilling

THE LANCET Gastroenterology & Hepatology

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ARTICLES • Volume 10, Issue 10, P896-903, October 2025

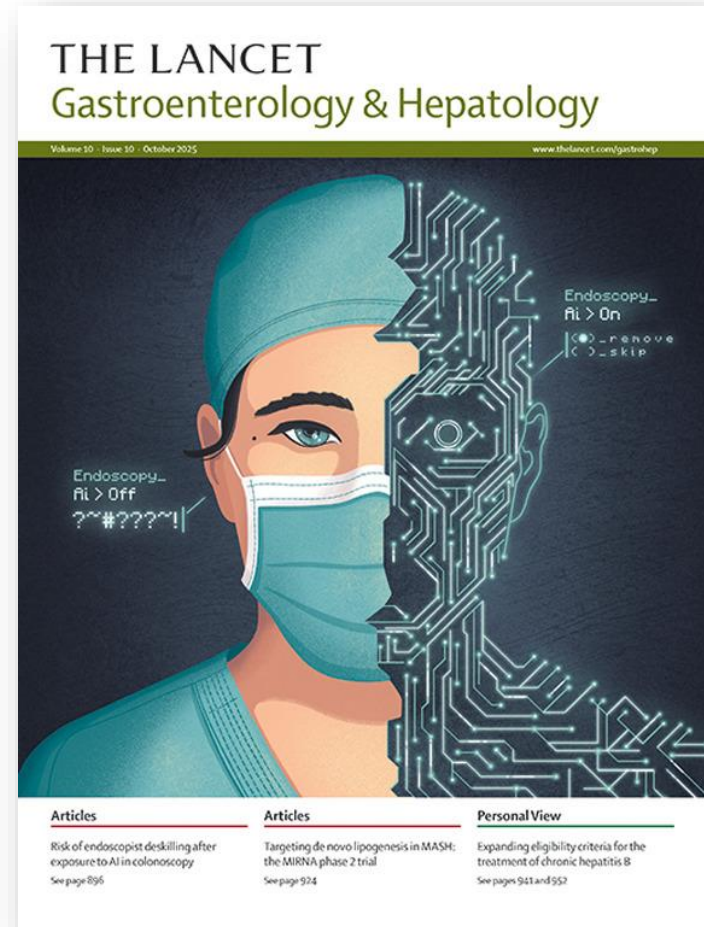
[Download Full Issue](#)

Endoscopist deskilling risk after exposure to artificial intelligence in colonoscopy: a multicentre, observational study

[Krzysztof Budzyń, MD](#)^{a,b} · [Marcin Romańczyk, MD](#)^{a,b}  · [Diana Kitala, PhD](#)^c · [Paweł Kołodziej, MD](#)^d · [Marek Bugajski, MD](#)^e · [Hans O Adami, MD](#)^{f,g} · et al. [Show more](#)

[Affiliations & Notes](#)  [Article Info](#)  [Linked Articles \(7\)](#) 

Adenoma detection fell 28% → 22% when AI was removed



Risk 2 — Cognitive offloading

- We offload thinking to the tool: convenient now, atrophy later
- Worse under time pressure — exactly the clinical scenario
- We must train the JUDGMENT of when NOT to offload the decision: high acuity, uncertainty, hands, time pressure

Article AI Tools in Society: Impacts on Cognitive Offloading and the Future of Critical Thinking

Michael Gerlich

Center for Strategic Corporate Foresight and Sustainability, SBS Swiss Business School, 8302 Kloten-Zurich, Switzerland; michael.gerlich@cantab.net

Abstract: The proliferation of artificial intelligence (AI) tools has transformed numerous aspects of daily life, yet its impact on critical thinking remains underexplored. This study investigates the relationship between AI tool usage and critical thinking skills, focusing on cognitive offloading as a mediating factor. Utilising a mixed-method approach, we conducted surveys and in-depth interviews with 666 participants across diverse age groups and educational backgrounds. Quantitative data were analysed using ANOVA and correlation analysis, while qualitative insights were obtained through thematic analysis of interview transcripts. **The findings revealed a significant negative correlation between frequent AI tool usage and critical thinking abilities, mediated by increased cognitive offloading.** Younger participants exhibited higher dependence on AI tools and lower critical thinking scores compared to older participants. Furthermore, higher educational attainment was associated with better critical thinking skills, regardless of AI usage. These results have implications for education and research.

Trends in Cognitive Sciences
A Cell Press journal

REVIEW · Volume 20, Issue 9, P676-688, September 2016 [Download Full Issue](#)

Cognitive Offloading

Evan F. Risko¹ · Sam J. Gilbert²

[Affiliations & Notes](#) [Article Info](#)

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» Trends

Physical action is sometimes used to reduce the cognitive demands of a task. This is known as cognitive offloading. Recent studies have begun to investigate the processes that trigger cognitive offloading, and the cognitive consequences of this behavior. Propensity to offload cognition is influenced by the internal cognitive demands that would otherwise be necessary. It is also influenced by metacognitive evaluations of our mental abilities. These metacognitive evaluations are potentially erroneous, which may lead to suboptimal offloading behavior.

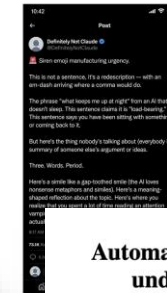
Choosing to Stay Human

...means choosing when and how to use AI.

ETHAN MOLLICK
MAY 26, 2026

775 81 89

If you go to your favorite social media site, you will find it full of posts that look suspiciously similar to each other:



Automation Bias in AI-Assisted Medical Decision-Making under Time Pressure in Computational Pathology

Emely Rosbach¹, Jonathan Ganz¹, Jonas Ammeling¹, Andreas Riener¹, and Marc Aubreville²

¹Technische Hochschule Ingolstadt, Ingolstadt, Germany

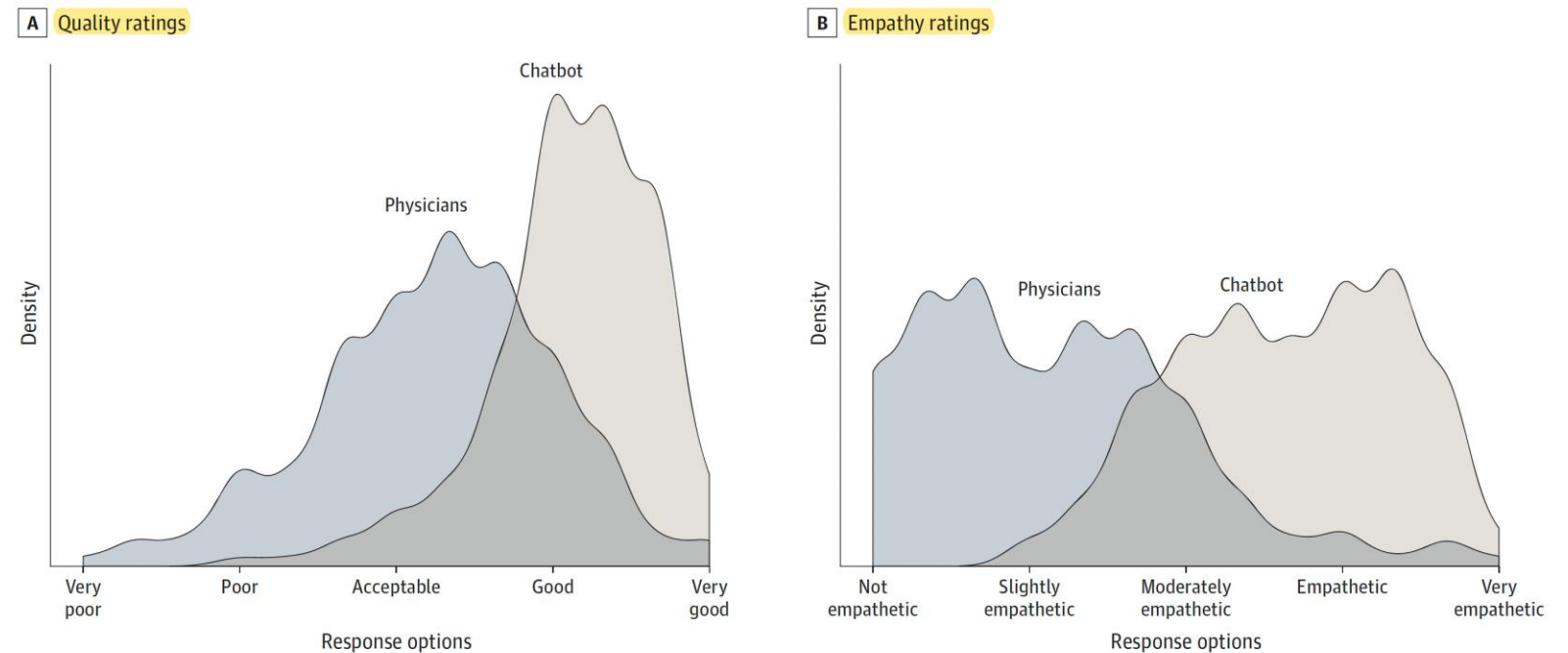
²Flensburg University of Applied Sciences, Flensburg, Germany
emely.rosbach@thi.de

Abstract. Artificial intelligence (AI)-based clinical decision support systems (CDSS) promise to enhance diagnostic accuracy and efficiency in computational pathology. However, human-AI collaboration might introduce automation bias, where users uncritically follow automated cues. This bias may worsen when time pressure strains practitioners' cognitive resources. We quantified automation bias by measuring the adoption of negative system consultations and examined the role of time pressure in a web-based experiment, where trained pathology experts (n=28) estimated tumor cell percentages. Our results indicate that while AI integration led to a statistically significant increase in overall performance, it also resulted in a 7% automation bias rate, where initially correct evaluations were overturned by erroneous AI advice. Conversely, time pressure did not exacerbate automation bias occurrence, but appeared to increase its severity, evidenced by heightened reliance on the system's negative consultations and subsequent performance decline. These findings highlight potential risks of AI use in healthcare.

We must train the more human skills

- Routine delegated to AI/agents; the doctor moves up to complex cases and supervision
- The premium is on the 'human': empathy, communication, shared decision-making
- More teamwork

Figure. Distribution of Average Quality and Empathy Ratings for Chatbot and Physician Responses to Patient Questions



What we'll do in this course and how

- **Studio, not show:** you do the work, we facilitate (predict-then-run)
- **The LLM as an object of critique:** the flagship exercise is catching the copilot's error
- **Intuition, not formulas:** maths as the key to a mystery (how an LLM knows heart attack $\approx$ MI)
- **A real Project:** build a clinical model with AI, and critique it (the critique is what counts)

Fluency is not truth.

A copilot writes code that runs, looks right... But it might be wrong. Your job is to catch it.

Why now

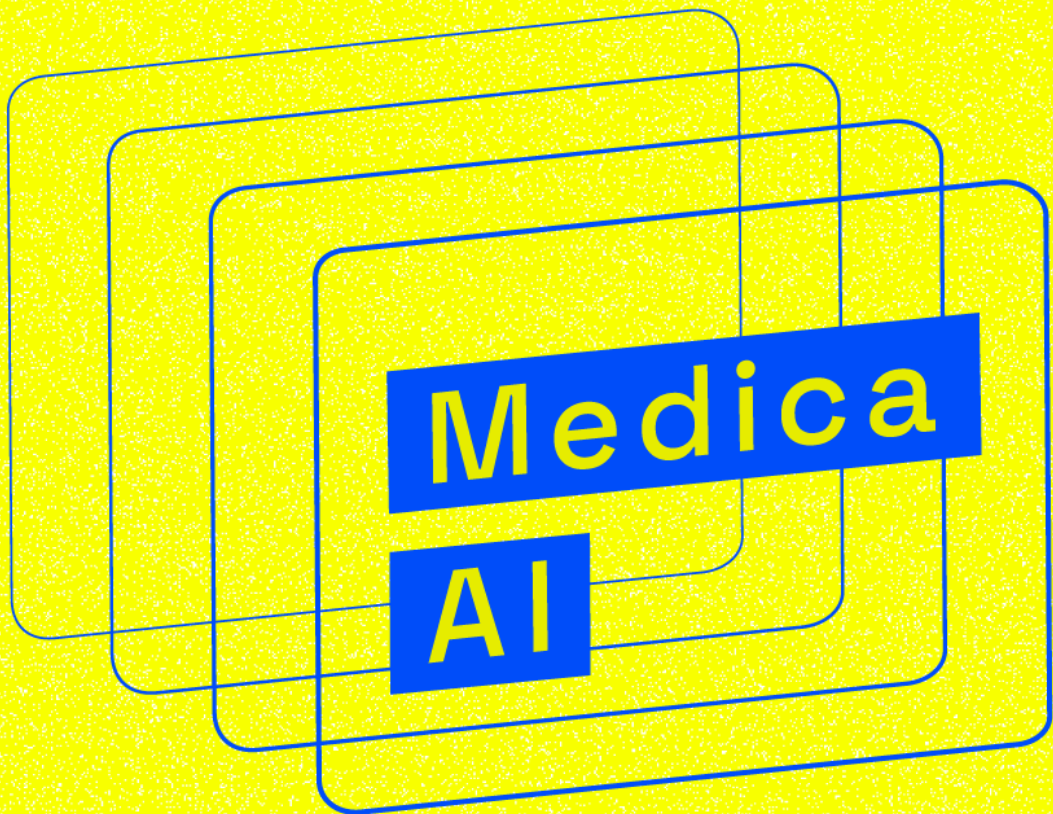
- You don't need to become an engineer
- You need to be the doctor who can read, verify and catch the copilot's error
- Critical judgment compounds: starting in 2nd year gives you years of advantage

Calendar for the course

IMLH (3.ª edição) — Calendário das sessões

Início 23 jun 2026 · 3ª/4ª/5ª × 4 semanas · materiais em inglês · Terças 3h (10–13h), Quartas/Quintas 4h (9h30–13h30)

Sessão	Semana	Dia	Data	Horário	Tema	Conteúdos
1	1	Ter	23 jun	10h–13h	Visão + Setup + Python I	Visão da cadeira · setup do Colab · Python (variáveis, listas)
2	1	Qua	24 jun	9h30–13h30	Python II	Mais Python (loops, funções)
3	1	Qui	25 jun	9h30–13h30	Python III + Pandas I	Ainda mais Python (dicionários) · Pandas (panda frames)
4	2	Ter	30 jun	10h–13h	Pandas II + Machine Learning I (Supervised) + Intuição Matemática I + Lançar projeto	Pandas (visualização) · Linear regression · 'aprender = minimizar o erro' (loss curve, derivada, gradient descent, local minima) · lançar o projeto final + formar grupos (3 grupos: 2 com 3 elements + 1 com 4 elementos; projeto igual para todos)
5	2	Qua	1 jul	9h30–13h30	Machine Learning II (Supervised) + Avaliação modelos	Logistic regression · Confusion matrix · sensitivity/specificity · recall/precision
6	2	Qui	2 jul	9h30–13h30	Generalização e afinação modelos + Machine Learning III (Supervised)	Overfitting e generalização · treino/teste · árvores de decisão · random forests · grid search
7	3	Ter	7 jul	10h–13h	Machine Learning IV (Non-supervised)	Clustering (k-means) · fenotipagem/segmentação de doentes
8	3	Qua	8 jul	9h30–13h30	Intuição Matemática II + Embeddings	Vectores · cosine similarity · Matriz como tabela e como transformação · embeddings
9	3	Qui	9 jul	9h30–13h30	Deep learning (intuição) + LLMs + RAG	Neurónio = soma + non-linearity · camadas = transformações · LLMs · RAG
10	4	Ter	14 jul	10h–13h	Responsible AI	YMYL · Viés · validação · explicabilidade · human-in-the-loop · dispositivo médico e AI Act (níveis de risco)
11	4	Qua	15 jul	9h30–13h30	Apresentações de projeto	Grupos apresentam projectos: Parte 1 (EDA) + Parte 2 (ML model, avaliação, impacto clínico, Responsible AI)
12	4	Qui	16 jul	9h30–13h30	Conferência	Medica AI



THE CONFERENCE 2026 _ Last Edition

IN CELEBRATION OF THE
AI APPRECIATION DAY

16 · 07 · 2026

Programme

07:00 Yoga Session

08:00 Registration and Welcome Coffee

Session 1 — The future of AI in healthcare

09:00 Opening Remarks
LEONOR BELEZA, PEDRO GOUVEIA
CHAMPALIMAUD FOUNDATION

09:20 ESSO - European Society Radiology special session
2050: Are you ready for AI driven healthcare?
REGINA BEETS-TAN MAASTRICHT UNIVERSITY,
MAASTRO AND EIBIR
PEDRO GOUVEIA CHAMPALIMAUD FOUNDATION

10:00 Healthcare's copernic moment
LUCIEN ENGELÉN TRANSFORMHEALTH

10:25 Healthcare after AI: Why human reinvention matters more than technology
NADIM HABIB NOVA SCHOOL OF BUSINESS AND ECONOMICS

10:45 COFFEE BREAK

Session 2 — Digital Hospitals

11:15 Digital Hospitals in China - Inside the world's fastest healthcare transformation
YIPENG SUN UNITED IMAGING HEALTHCARE

11:35 Medica AI FightClub – Digital hospitals
SPEAKER IN FAVOR – HENRIQUE MARTINS,
CHAMPALIMAUD FOUNDATION
SPEAKER AGAINST – ANTÓNIO VAZ CARNEIRO,
INSTITUTE FOR EVIDENCED BASED HEALTHCARE

12:20 AI, neuroplasticity and the reinvention of brain therapy
JOE PATON, JOHN KRAKAUER, MARCELO MENDONÇA
CHAMPALIMAUD FOUNDATION

12:40 From pixel to scalpel: building the AI and AR infrastructure for the future of medicine
JOÃO SANTINHA, TIAGO MARQUES
CHAMPALIMAUD FOUNDATION

13:00 Medica AI drone group picture
(Amphitheatre)

13:10 LUNCH BREAK

Session 3 — AI in healthcare education

14:30 Digital Academy by Lisbon School of Medicine
JOÃO EURICO FONSECA, MARIA BATRAKOVA
LISBON SCHOOL OF MEDICINE,
CHAMPALIMAUD FOUNDATION

14:50 Google AI Studio: How vibe coding is rewriting the rules in healthcare
JOANA CARRASQUEIRA GOOGLE DEEPMIND

15:10 Ctrl + Alt + Cure: Students building medical AI!
IARA CALDEIRA, FABIANA LU, MARIA PINHEIRO
LISBON SCHOOL OF MEDICINE

15:20 Google MEDICA AI Hackathon winner pitch

15:30 Book Launch: Artificial Intelligence and Medicine
PAULO SIMÕES SOUTHERN REGION OF THE PORTUGUESE MEDICAL ASSOCIATION

15:40 Dogma AI
TUPAC MARTIR SATORE STUDIO,
MARKUS MAURER CHAMPALIMAUD FOUNDATION

16:00 COFFEE BREAK

Session 4 — From research to clinical impact

16:30 What the body whispers: the future of brain health through everyday signals
IGOR MATIAS UNIVERSITY OF GENEVA

16:50 AI Care: Putting clinical outcomes at the center?
CATARINA FARINHA SWORD HEALTH

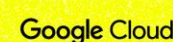
17:10 Why AI will redesign every patient journey
EDUARDO FREIRE RODRIGUES UPHILL

17:30 Treating risk before breast cancer exists: The new age of prevention
ANDREA DE CENSI CHAMPALIMAUD FOUNDATION

17:50 Champions of Change - Special Session

18:10 2026 Breast Cancer Research Program Awards
MARIA JOÃO CARDOSO CHAMPALIMAUD FOUNDATION

18:30 SUNSET



Doctors who shape AI, not doctors shaped by it.

It starts today.